

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
group_sec: 10
timesteps_orig: 30
timesteps: 3
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
forecast_steps: 20

LSTM Multi-Step Training Report

Report Generated: 2025-05-10 09:23:11

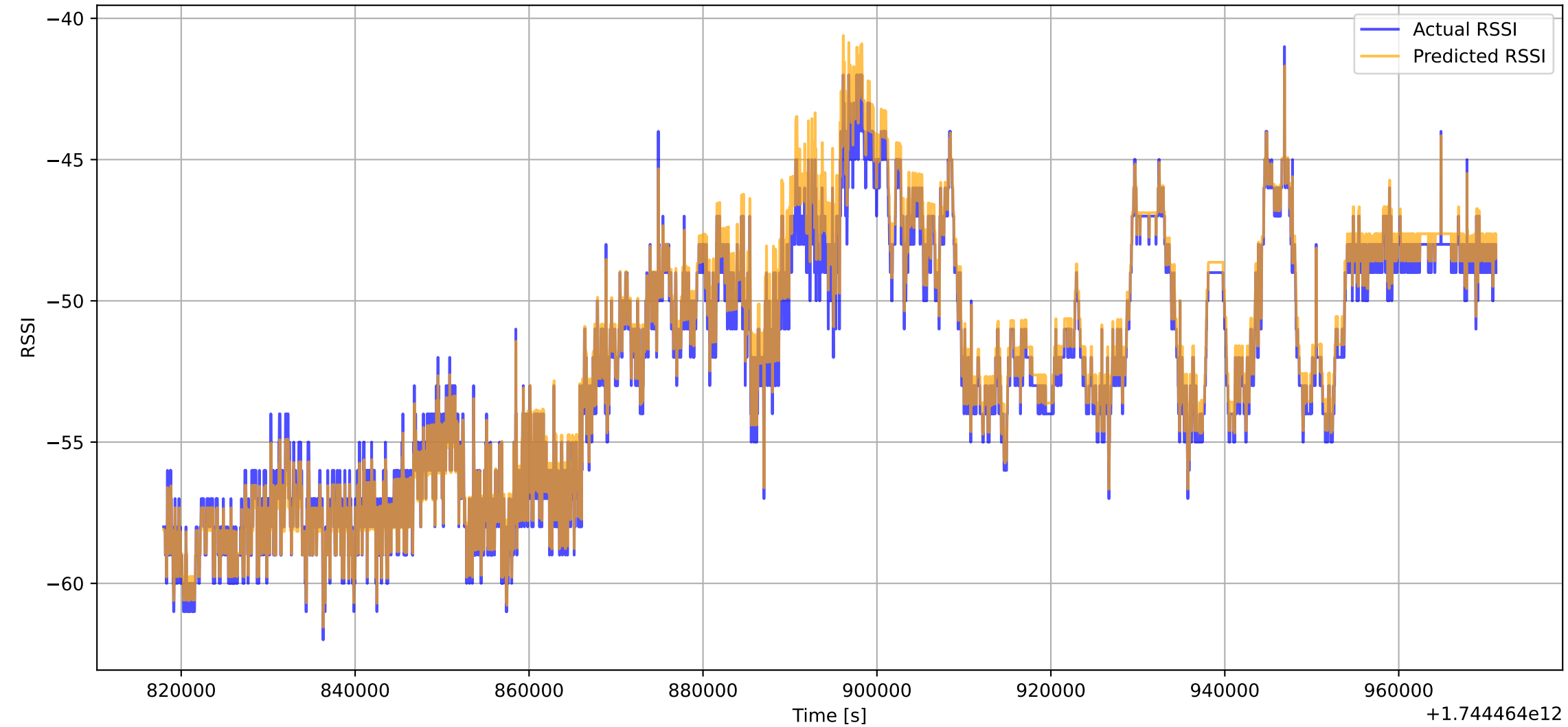
Training Start: 2025-05-10_08-52-59
Training End: 2025-05-10_08-57-51
Training Duration: 4.87 minutes

Best Hyperparameters:
Number of Layers: 1
Activation: elu
Optimizer: Nadam
Dropout Rate: 0.20

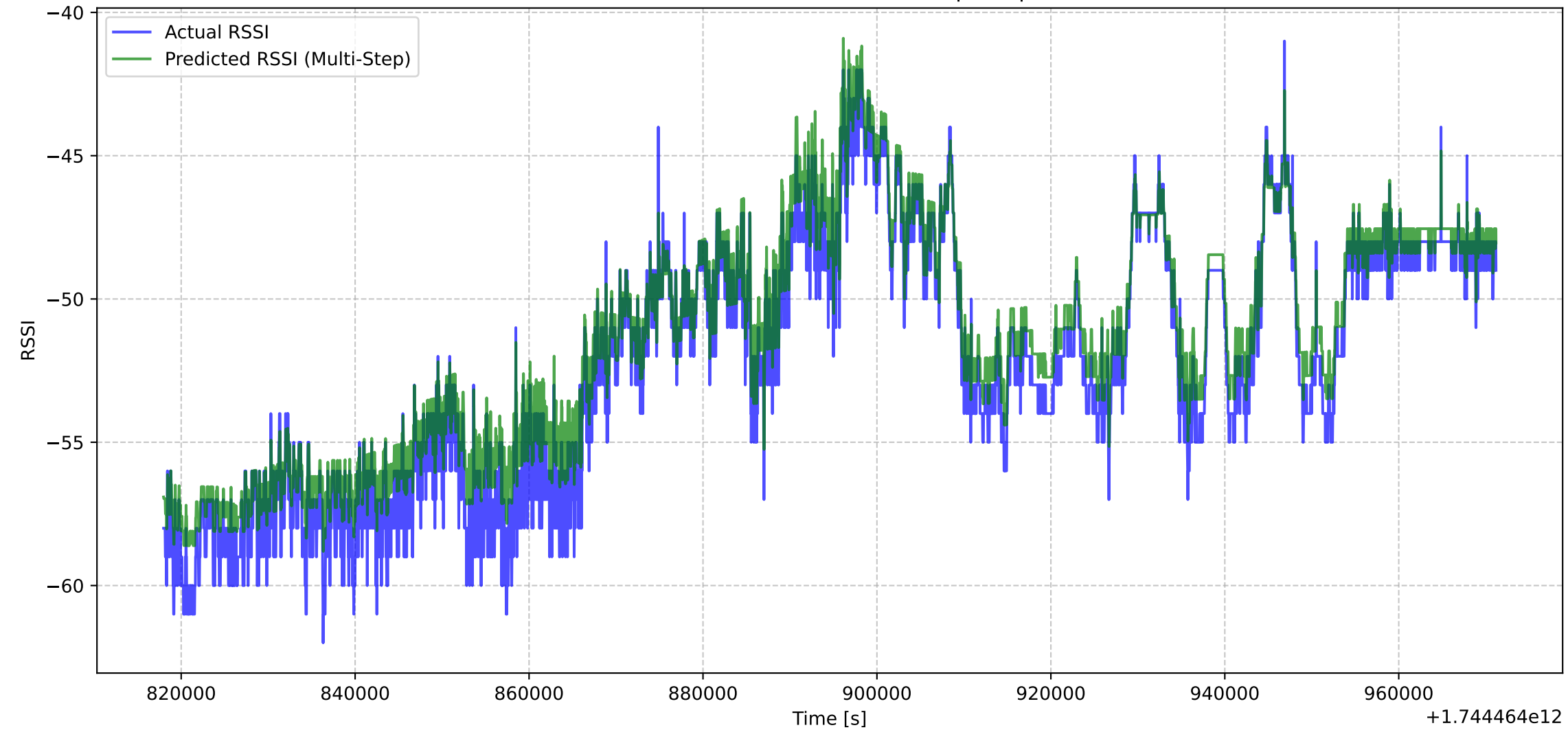
MAE: 1.0786
MSE: 1.9599
RMSE: 1.4000
R2 Score: 0.8921

Multi-Step Forecast:
Forecast Steps: 20
MAE: 0.0083
MSE: 0.0001
RMSE: 0.0111
R2 Score: 0.9694

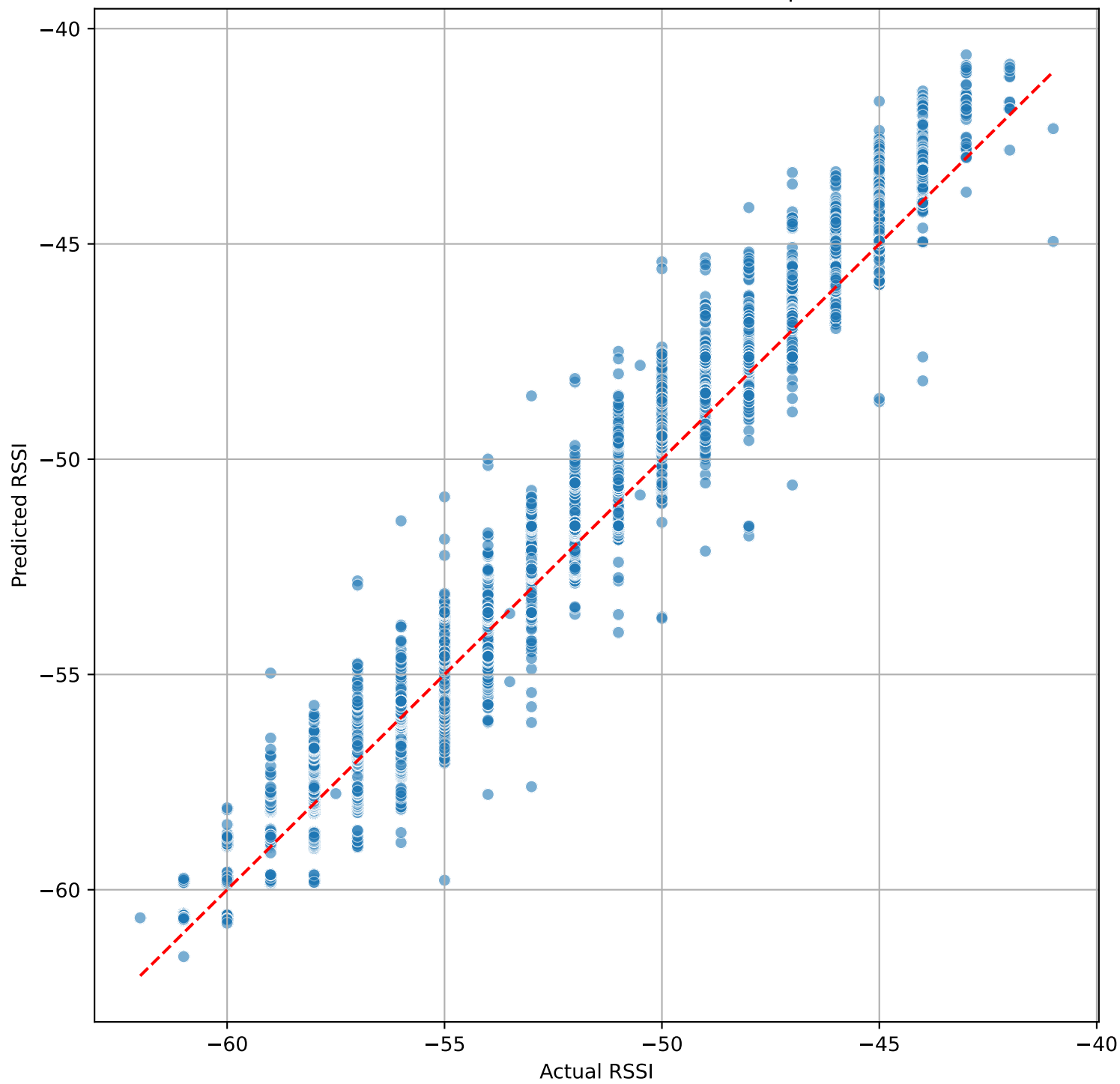
Actual vs Predicted RSSI (First Step)



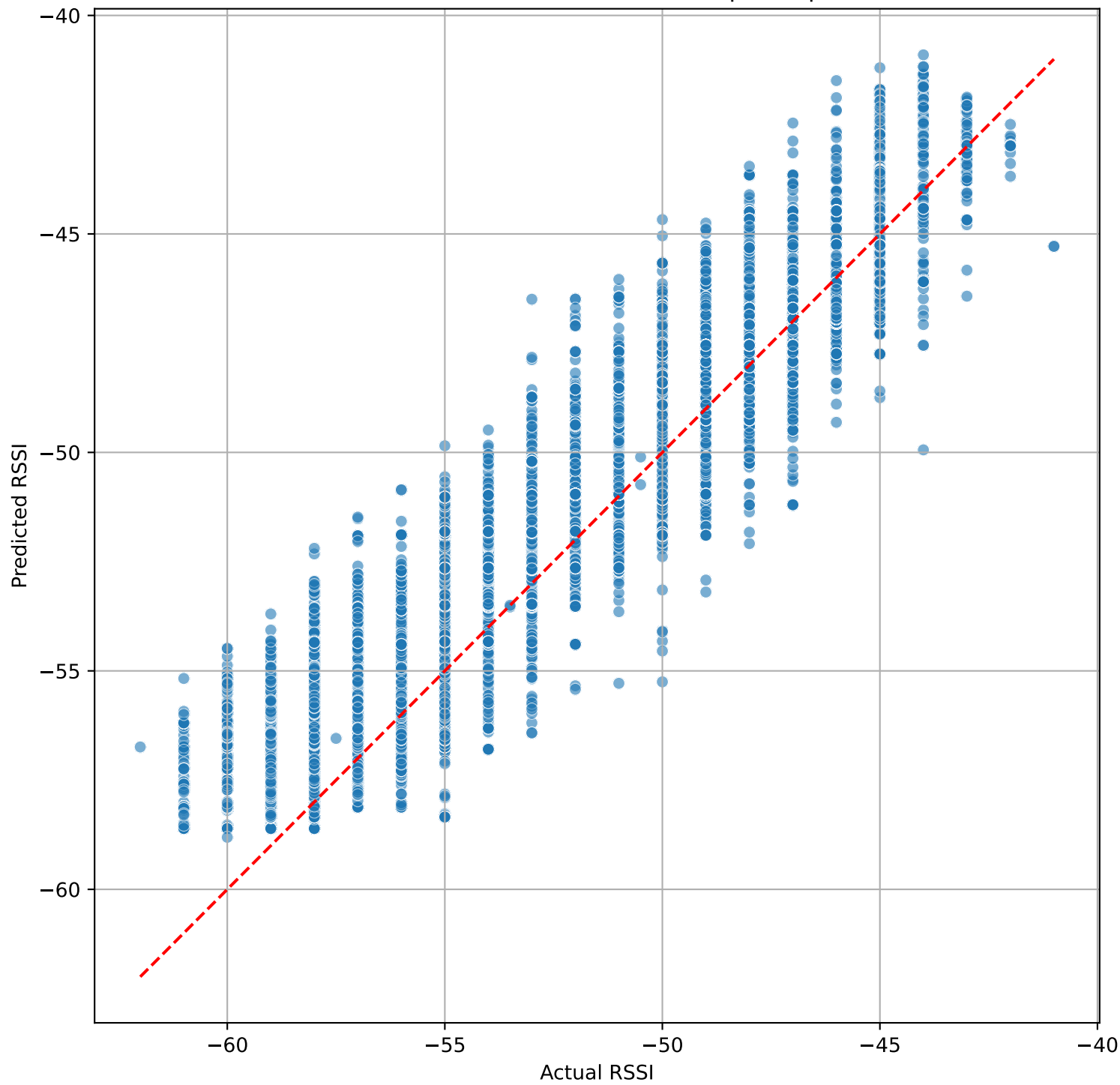
Actual vs Predicted RSSI (Multi-Step (step 20))



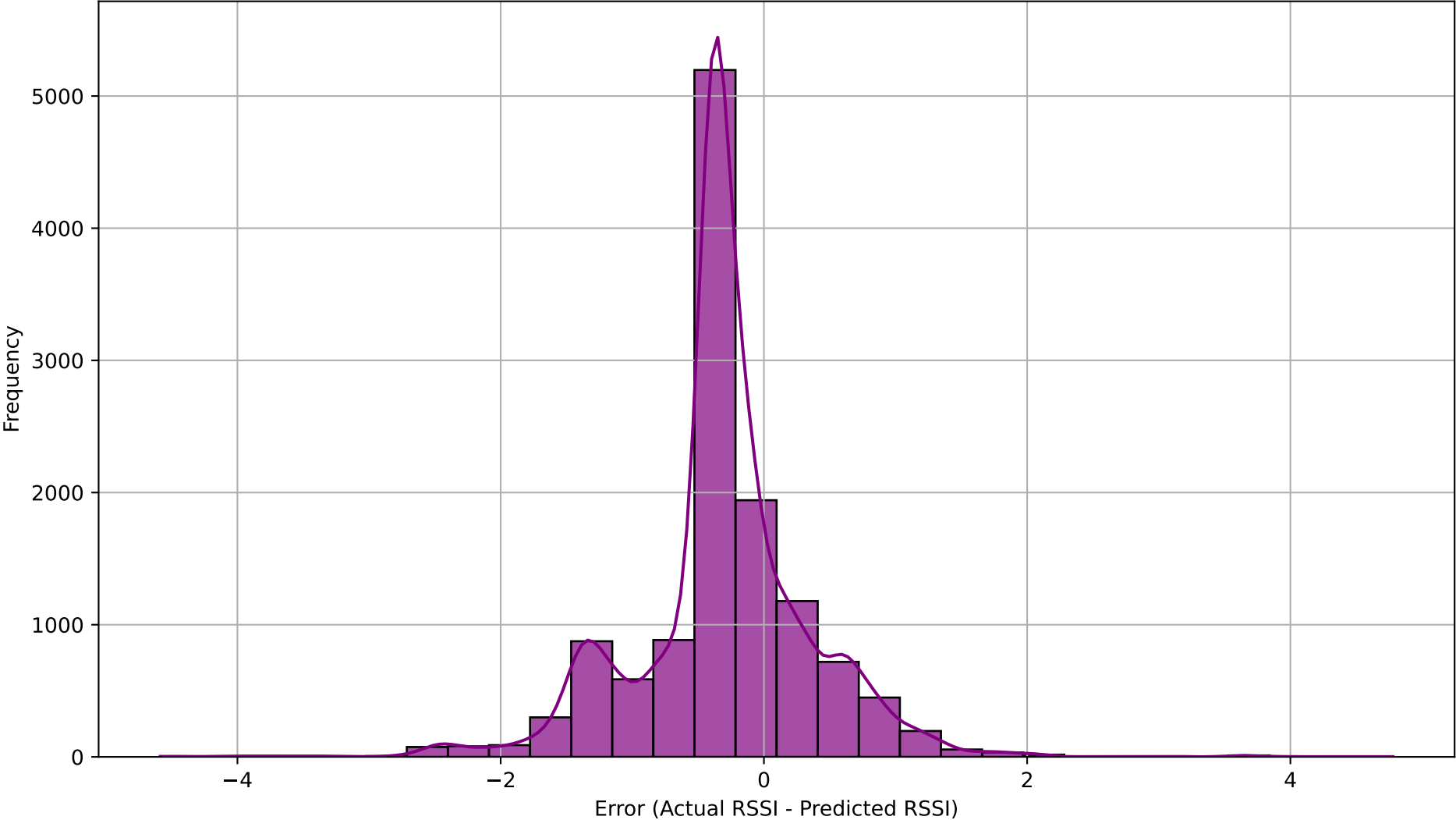
Actual vs Predicted RSSI (First Step)



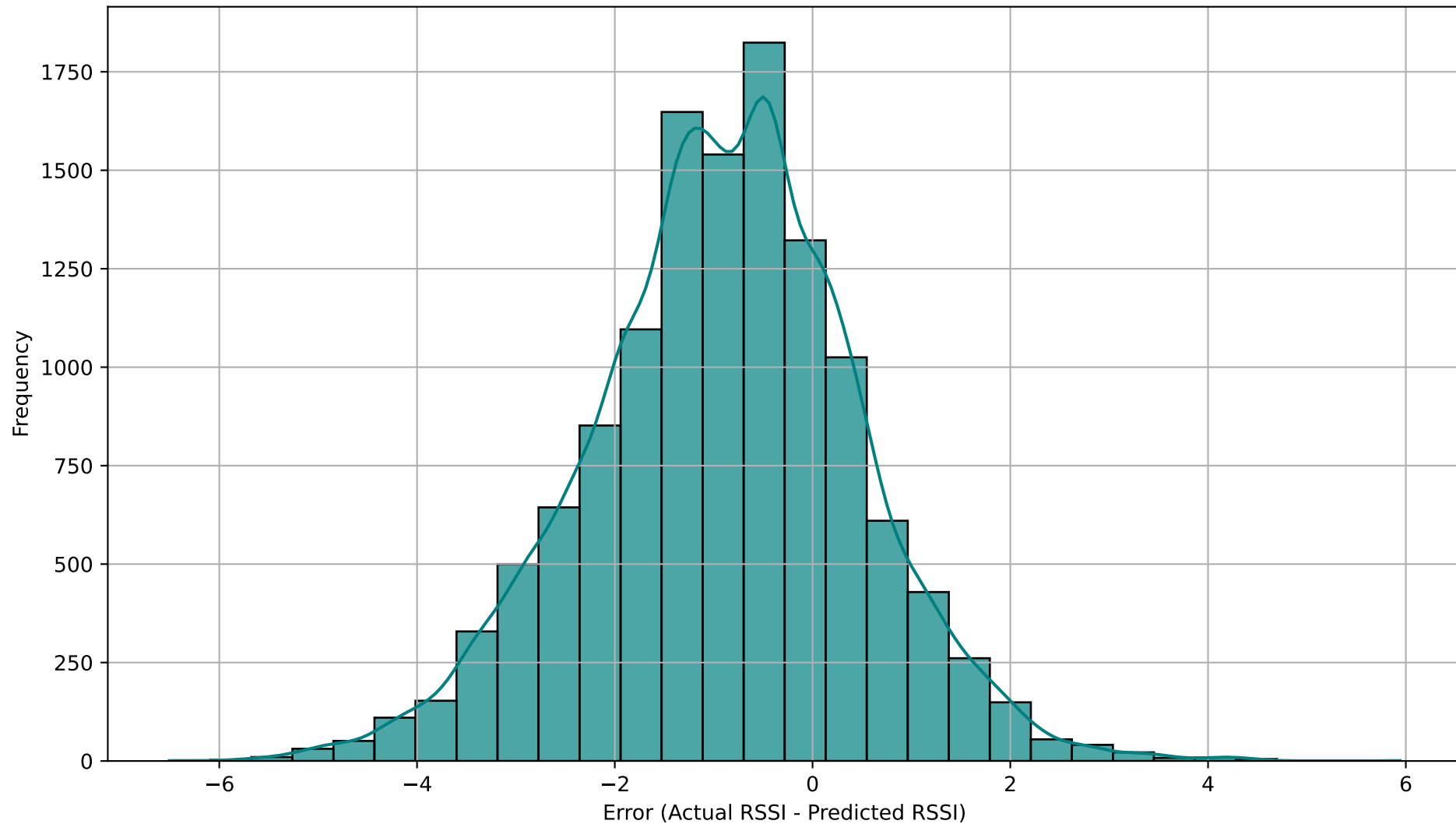
Actual vs Predicted RSSI (Last Step - step 20)



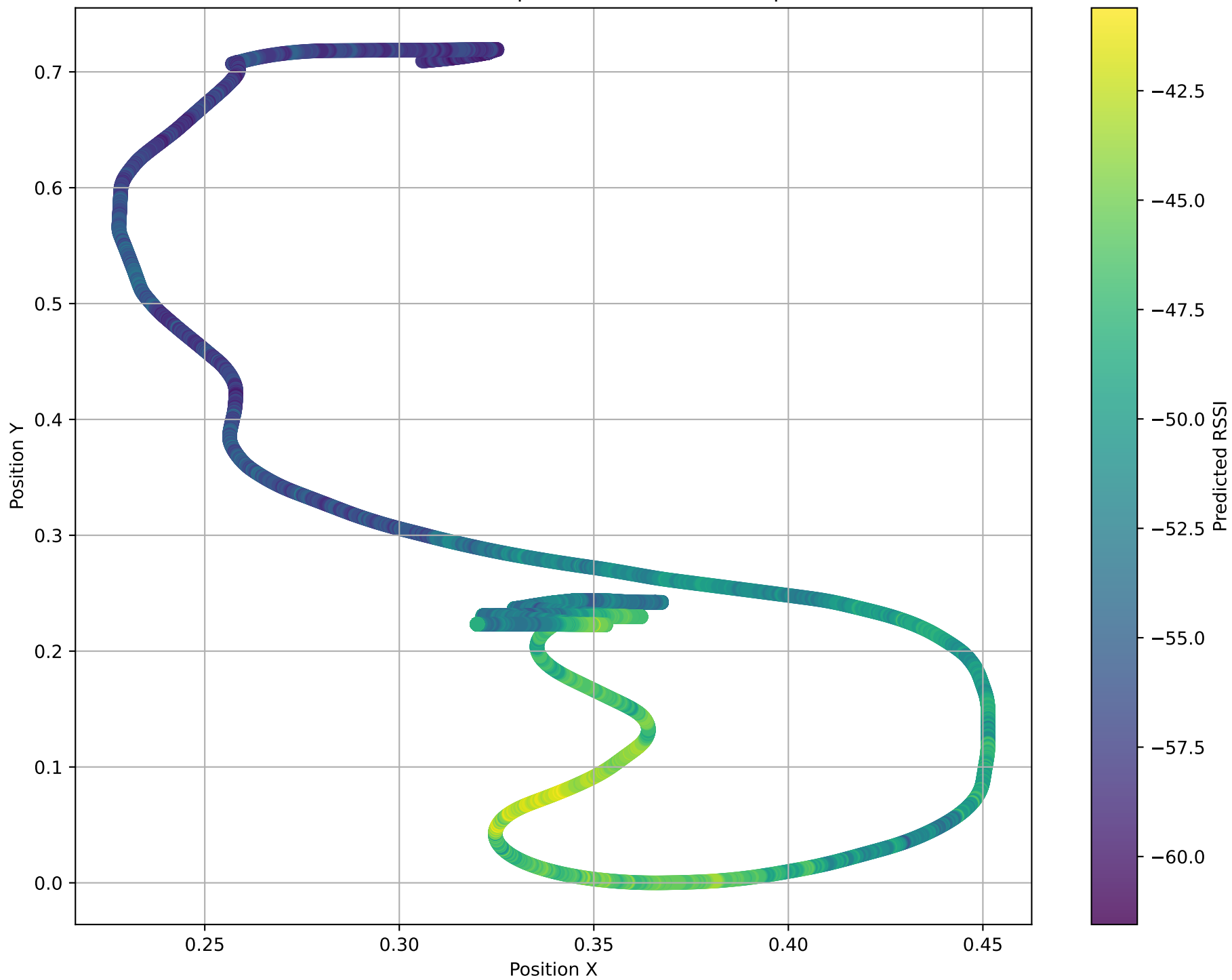
Distribution of Prediction Errors (First Step)



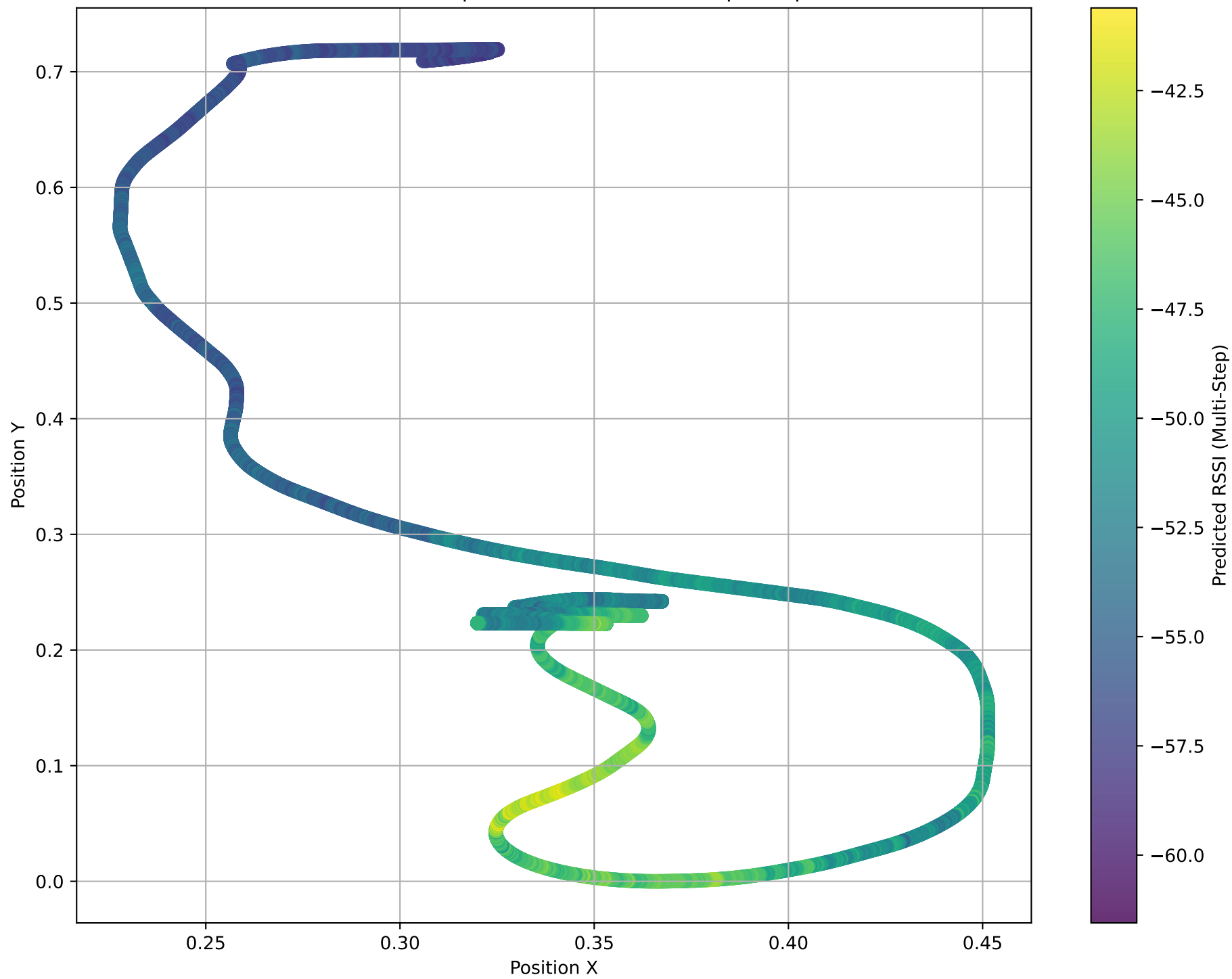
Distribution of Prediction Errors (Last Step - step 20)



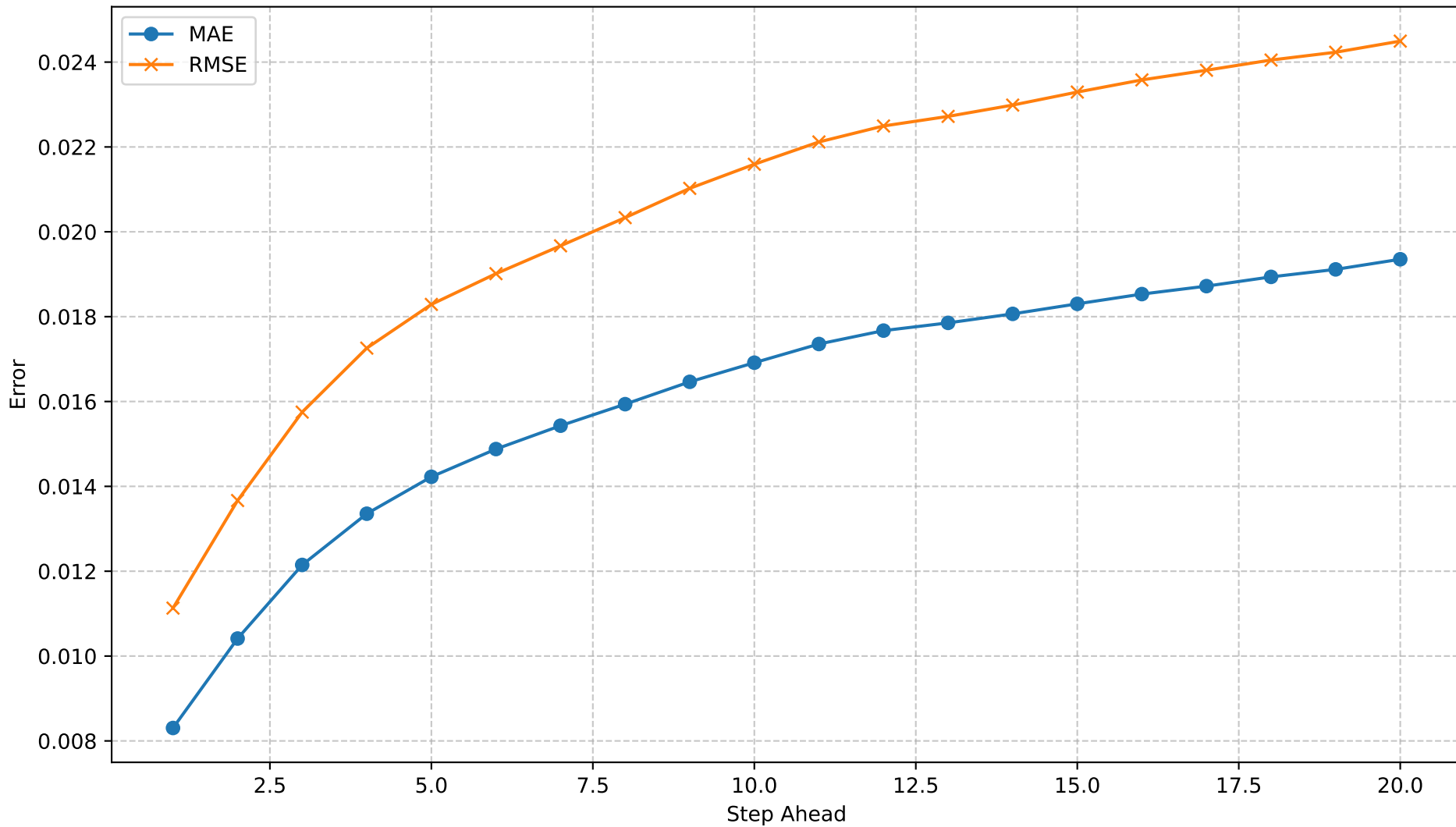
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet
Modelo: LSTM
Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.01 / 0.01
- Mediana : 0.02 / 0.02
- 20º passo : 0.02 / 0.02

Crescimento do erro: 133.0% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 20 (0.02), 19 (0.02), 18 (0.02)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.